

VRIJE
UNIVERSITEIT
BRUSSEL

Analog Electronics lab session

Arno, Chang-Shao, Ezgi, Xuepu

Session 3 : Design of differential pairs and of an OTA

In this exercise you will learn ...

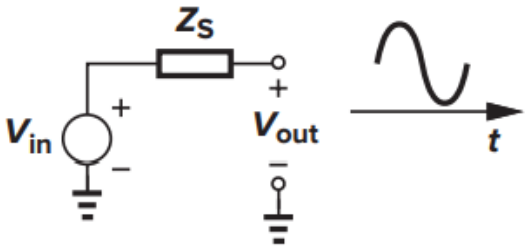
- Fundamentals about differential pairs
- How to "design on paper"
  - Differential pair with resistive load
  - OTA
- Learn to value $\frac{g_m}{I_D}$- and $\frac{g_m}{g_{ds}}$-plots !
  - with those plots one can identify design trade-offs!
- How to code in MATLAB the OTA-design

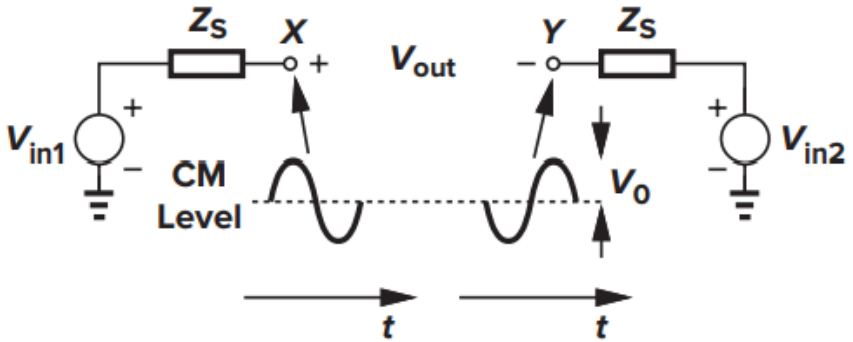
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What is single ended and differential operation?

Single Ended Signal: Signal measured with respect to a fixed potential, usually the ground.

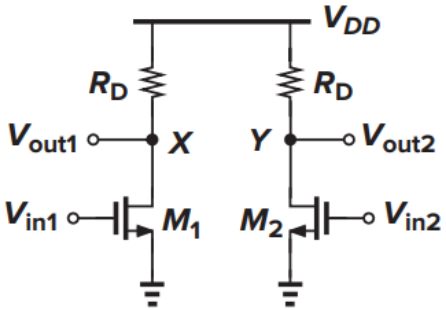
Differential Signal: Signal measured between two nodes that have equal and opposite deviations around a fixed potential.

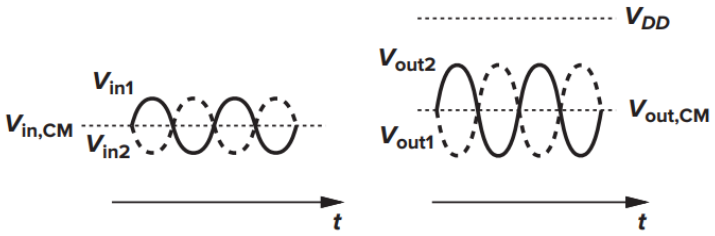




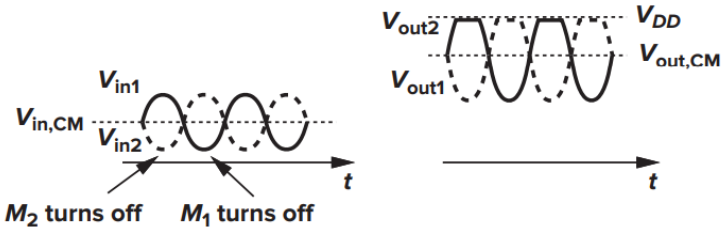
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Principles of a Basic Differential Pair





Normal Operation

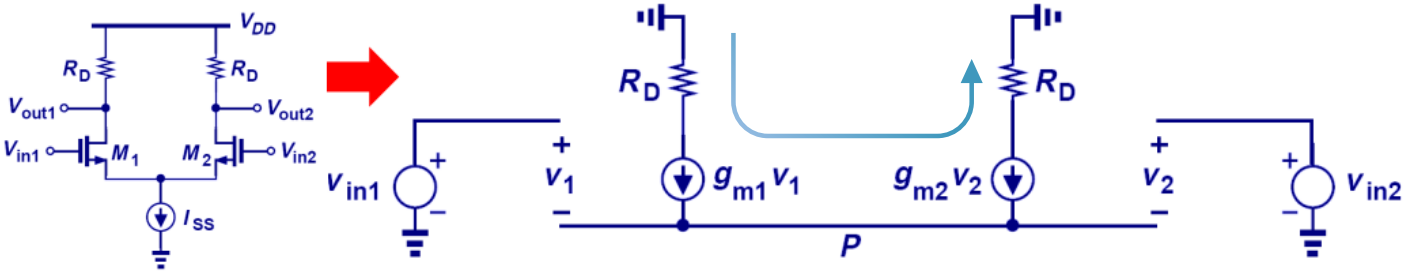


Input CM level is excessively low

Bias currents should have minimal dependence on input CM level!

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Small Signal of a simple Differential Amplifier and Virtual Ground Concept



1. KVL around the input networks
   $v_{in1} - v_1 = v_p = v_{in2} - v_2$

2. KCL at node P
   $gm_1 v_1 + gm_2 v_2 = 0$

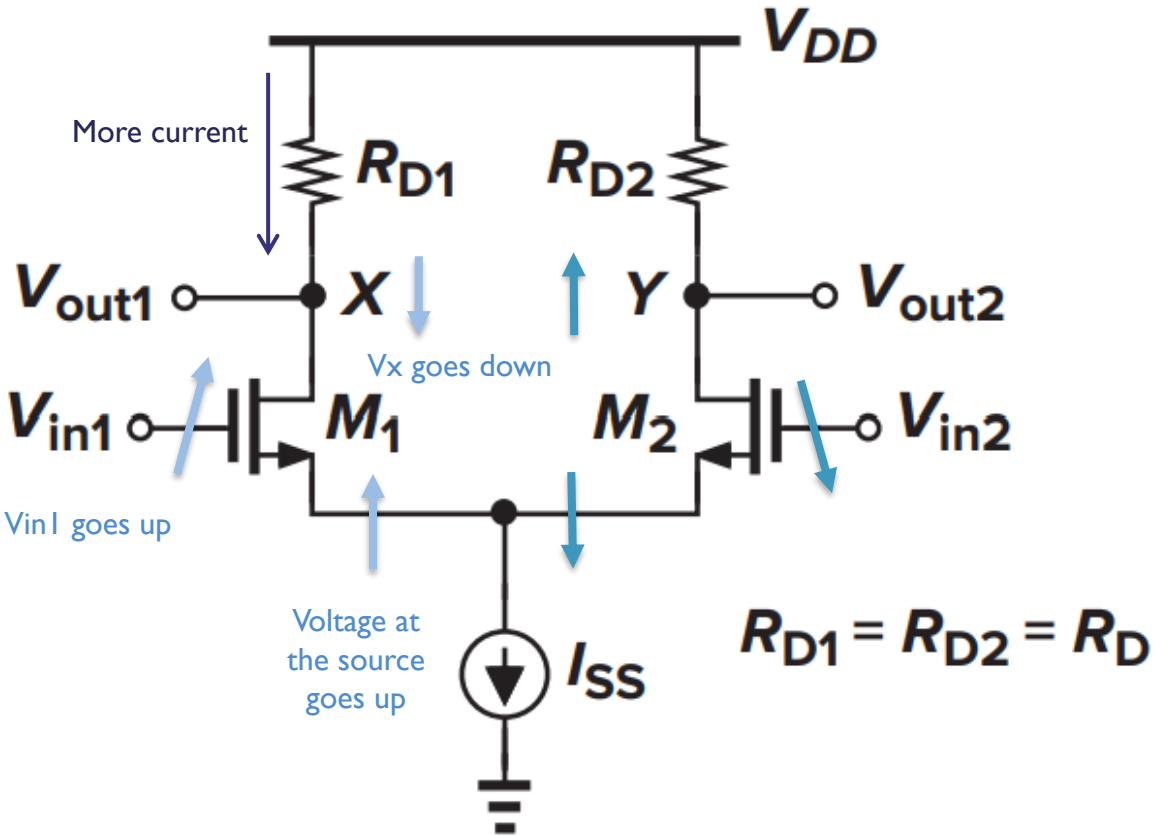
$$-v_1 = v_2$$

$$v_1 = v_{in1}$$

$$v_{in1} - v_1 = v_p = 0$$

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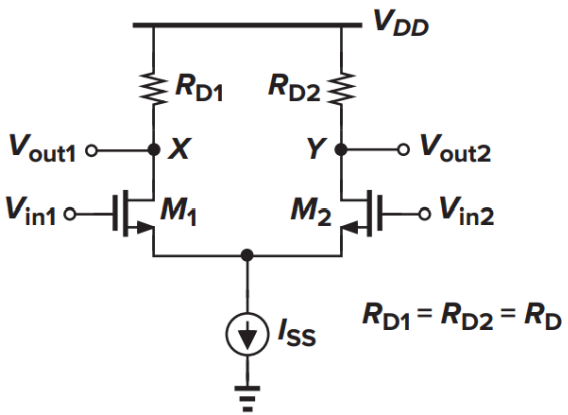
Half Circuit for Differential



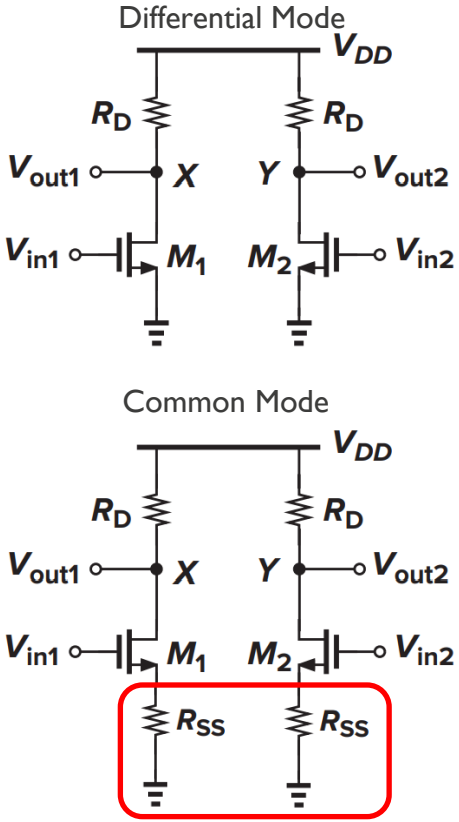
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Half Circuit for Differential and Common Mode

For Symmetric Circuits!



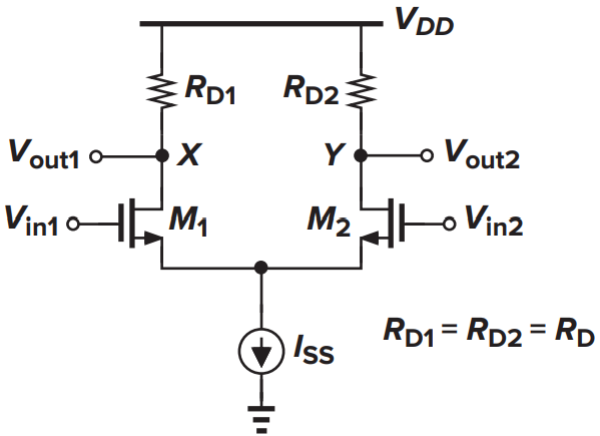




Hint for exercise 1a!!

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Large Signal



Think of what happens to nodes X and Y, when $V_{in1} - V_{in2}$ varies from -∞ to + ∞.

When:

- V_{in1} is much more negative than V_{in2}
- V_{in1} is brought closer to V_{in2}
- V_{in1} is more positive than V_{in2}
- V_{in1} is much more positive than V_{in2}

Hint for exercise 1b!!

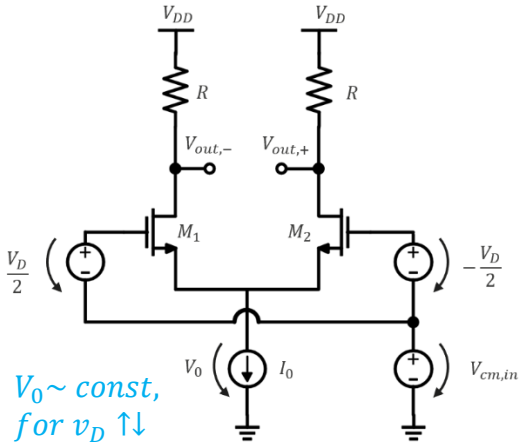
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- Solve exercise 1 a-d
  - ETA: 20 min

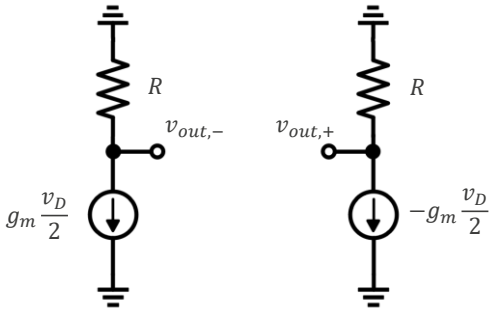
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Exercise 1

a) Give the equation for $A_{v,-} = \frac{v_{out,-}}{v_D}$, $A_{v,+} = \frac{v_{out,+}}{v_D}$ and $A_{v,diff}$?



small signal analysis



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Exercise 1.b

b) Draw the voltages V_{out1} and V_{out2} on Fig2 and the difference $V_{out1} - V_{out2}$ on Fig3 as $V_{in1} - V_{in2}$ varies from -∞ to + ∞. Indicate:

i. Upper and lower bound for these voltages
ii. The inversion level of M1 and M2
iii. Small signal voltage gain

Think of what happens to nodes X and Y, when $V_{in1} - V_{in2}$ varies from -∞ to + ∞. Start with the hint!

When:

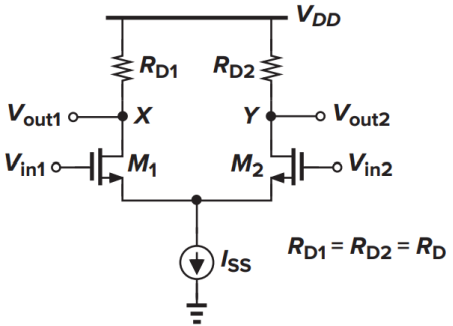
- V_{in1} is much more negative than V_{in2}
- V_{in1} is brought closer to V_{in2}
- V_{in1} is more positive than V_{in2}
- V_{in1} is much more positive than V_{in2}

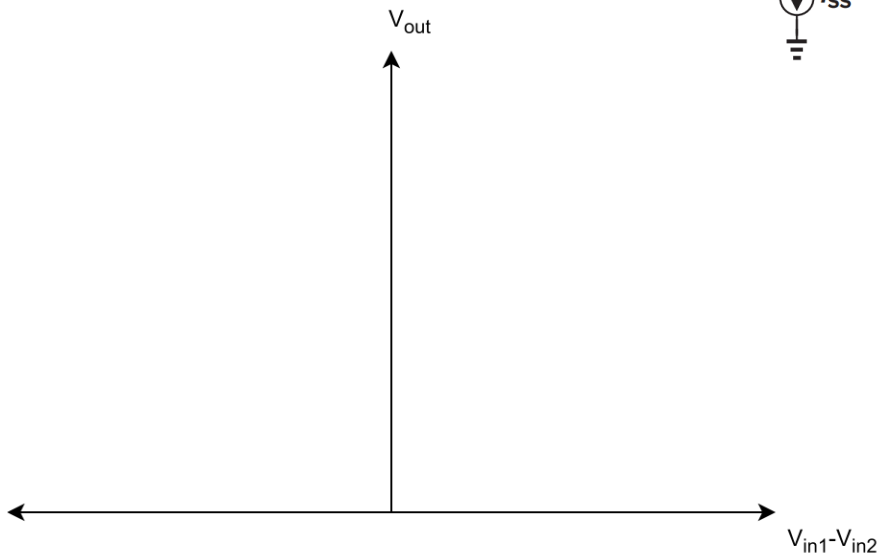
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Exercise 1.b

When:

- V_{in1} is much more negative than V_{in2}
- V_{in1} is brought closer to V_{in2}
- V_{in1} is more positive than V_{in2}
- V_{in1} is much more positive than V_{in2}





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Exercise 1

c) Give the amount of current, which is flowing through M_1 and M_2, when

(i) $V_D = 0$,

(ii) $V_D << V_{OV}$,

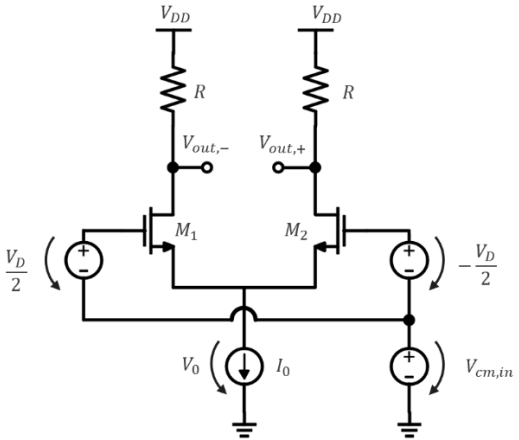
(iii) $V_D >> V_{OV}$.

c) $I_{D,M1}$ and $I_{D,M2}$ given as follows

i. $I_{D,M1} = I_{D,M2} =$

ii. $I_{D,M1} = I_0$ and $I_{D,M2} =$

iii. $I_{D,M1} = 0$ and $I_{D,M2} =$



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Exercise 1

d) Based on Fig. 3, derive an equation describing the relationship between V_{swing}, V_{OV} and A_V.
Remark: $V_{swing} = max(V_{out,+} - V_{out,-}) - min(V_{out,+} - V_{out,-})$

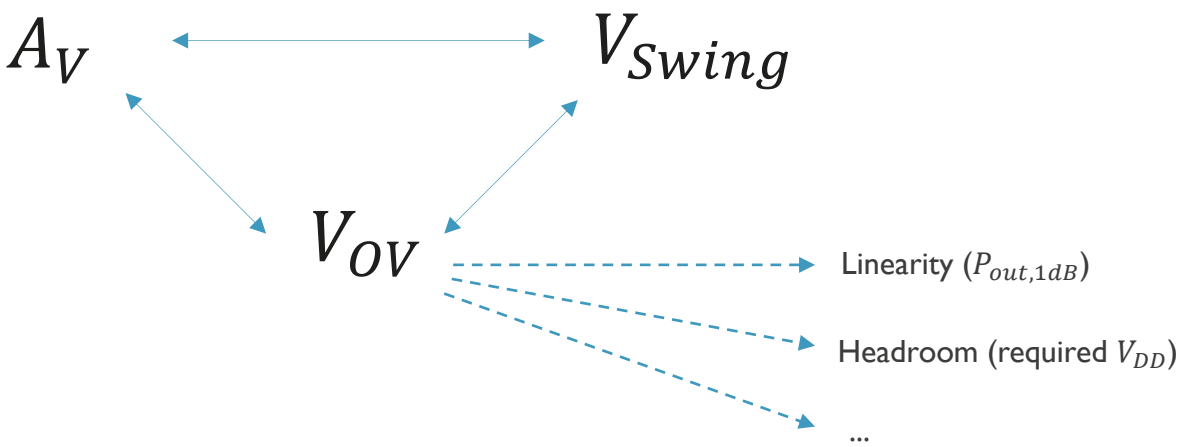
d) $\frac{g_{m_{1,2}}}{I_{D_{1,2}}} =$ We know $g_m = \frac{2I_D}{V_{ov}}$

We know $g_m R = A_{v,diff}$

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Summary

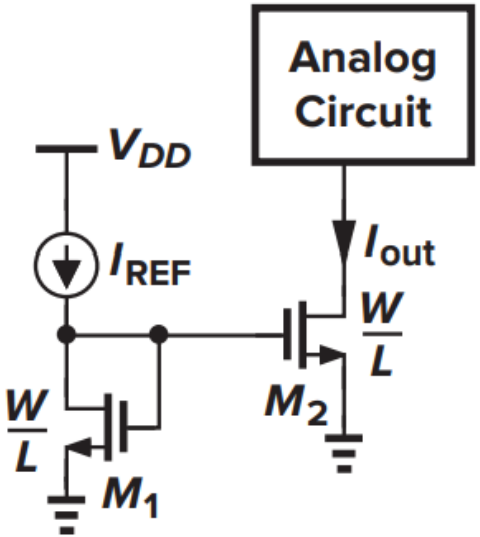
- Waveforms of an ideal differential amplifier has been presented
- (in this exercise) Trade-off triangle of differential amplifiers has been identified
  - ... in reality, one has even an octagon (see literature [1-3])



➢ Now, the differential amplifier is going to be designed

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Exercise 1



Current Mirror Architecture

$$I_{REF} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_{GS} - V_{TH})^2$$

$$I_{out} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_2 (V_{GS} - V_{TH})^2$$

$$I_{out} = \frac{(W/L)_2}{(W/L)_1} I_{REF}$$

Ignore channel length modulation.

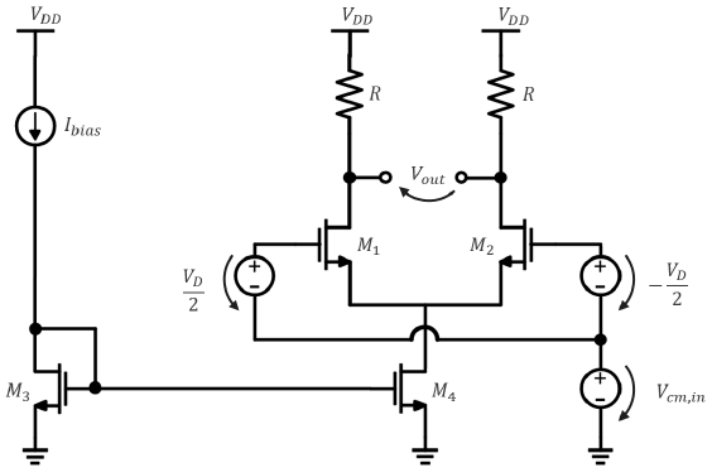
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- Solve exercise 1 e-h (option i&j will be solved together)
  - ETA: 20 min

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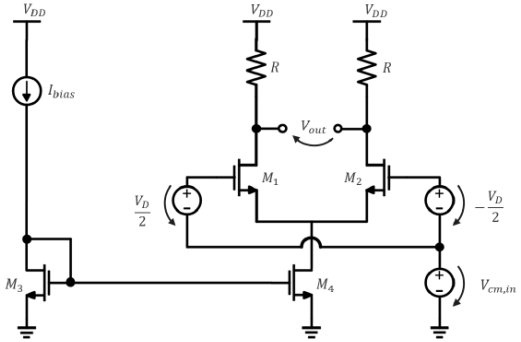
Exercise 1

e) Use the plots in option 1.b to give V_{swing} as a function of I_{bias}.

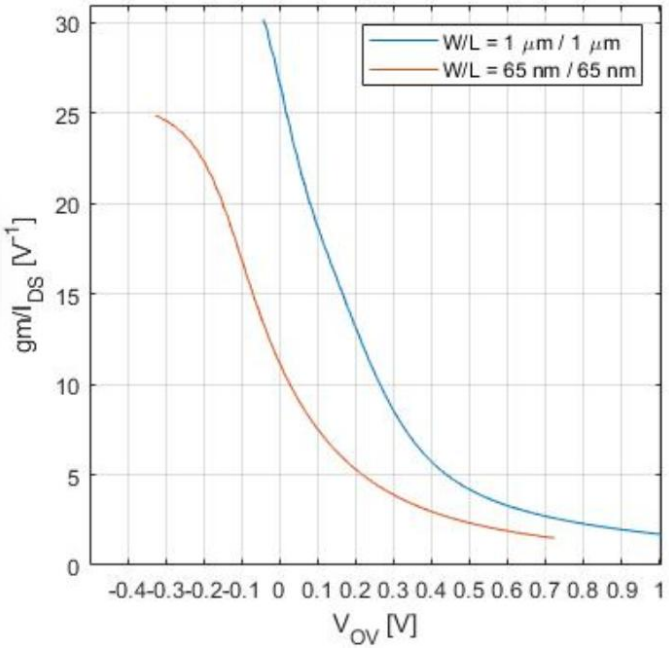


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Exercise 1



f) Calculate, what $\frac{g_m}{I_D}$ is required to have exactly $A_V = 0$ and $V_{swing} = 0.8V_{pp}$.



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Exercise 1

g) Calculate R and g_m such that this circuit passes exactly the V_{swing}-, P_{diss} and A_V-specification.

g) First, calculate the max. allowed current

$P_{diss} =$

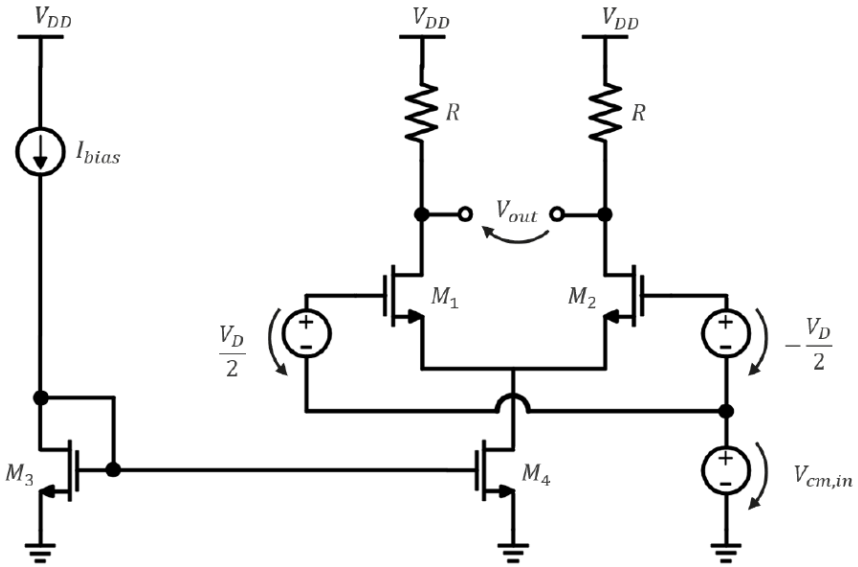
$\Leftrightarrow$

Then, calculate $g_{m_{1,2}}$ (from $\frac{g_{m_{1,2}}}{I_{D_{1,2}}}$ found previously)

$g_{m_{1,2}} =$

Now, calculate R

$\Leftrightarrow R =$

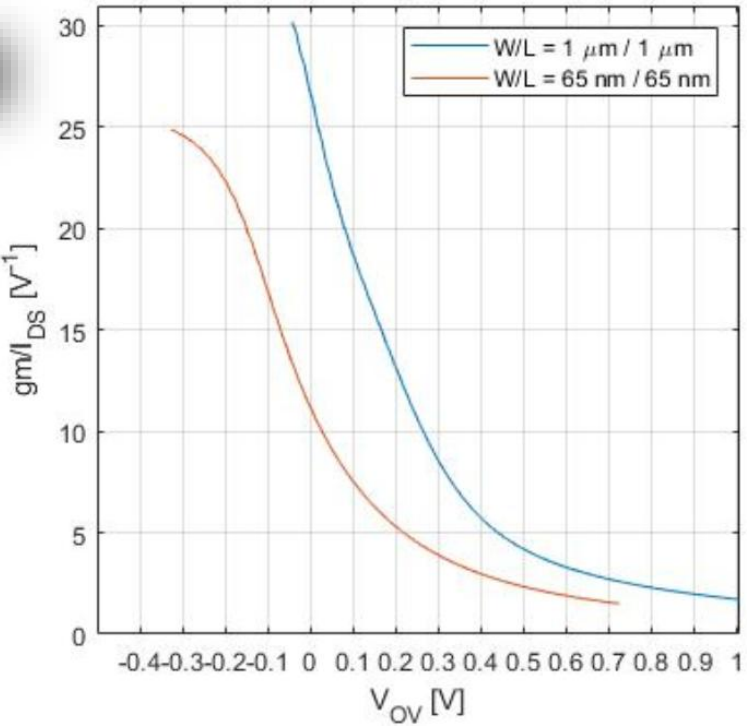


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Exercise 1

h) Calculate the required width W_1 and W_2 as well as L_1 and L_2 with aid of Fig. 5 and Fig. 6.

h) Reminder: $\frac{g_{m_{1,2}}}{I_{D_{1,2}}} = 5$

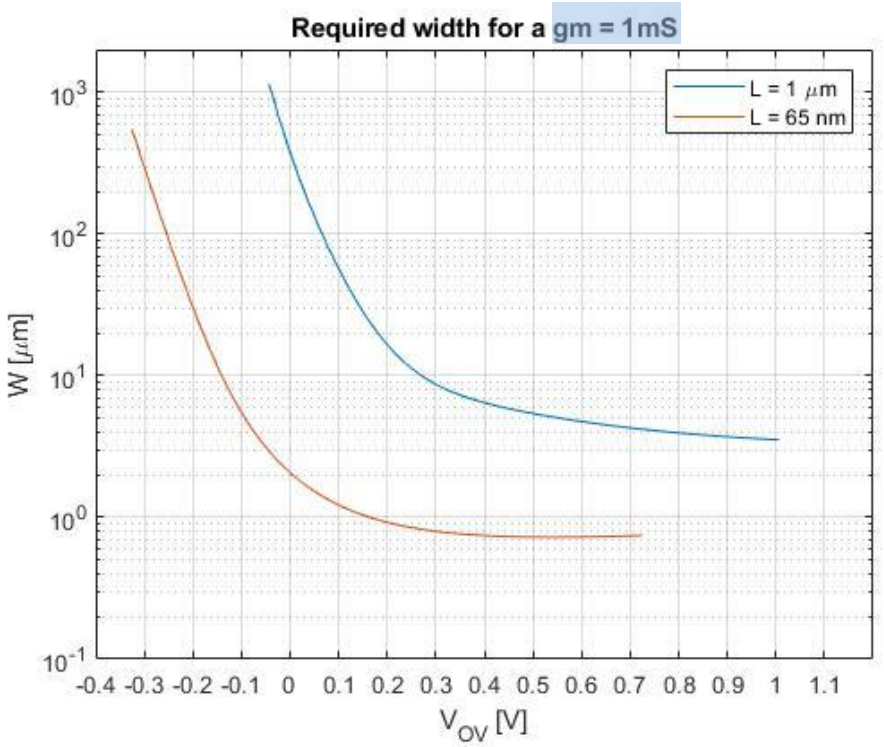


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Exercise 1

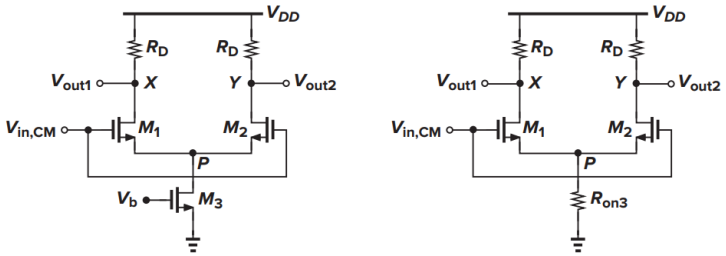
h) Now determine the width $W_{1,2}$ at $V_{OV} = 0.2V$

from g): $g_m =$



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Common Mode Analysis



What happens if $V_{in,CM}$ varies from 0 → V_{DD}

$V_{in,CM} = 0$

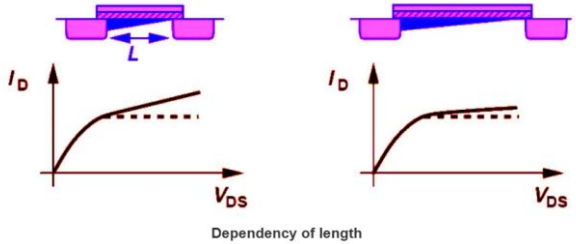
M1, M2 OFF → ID3 = 0

$V_{in,CM} > V_{th}$

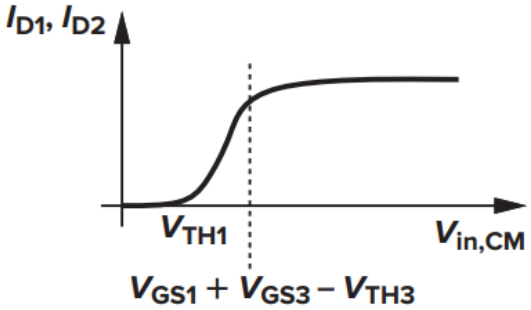
M1, M2 ON → ID3 increases

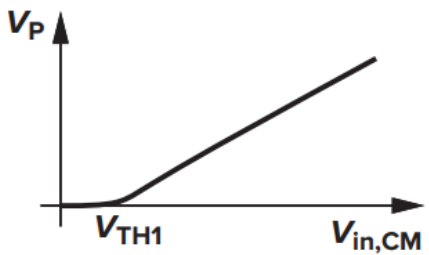
$V_{in,CM} > V_{GS1} + V_{GS3} - V_{th3}$ Saturation!

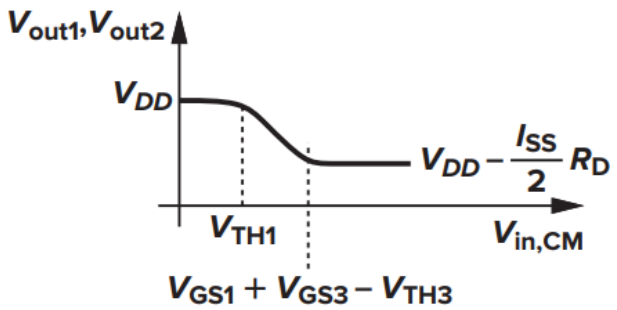
M1, M2 ON → ID3 constant



Dependency of length







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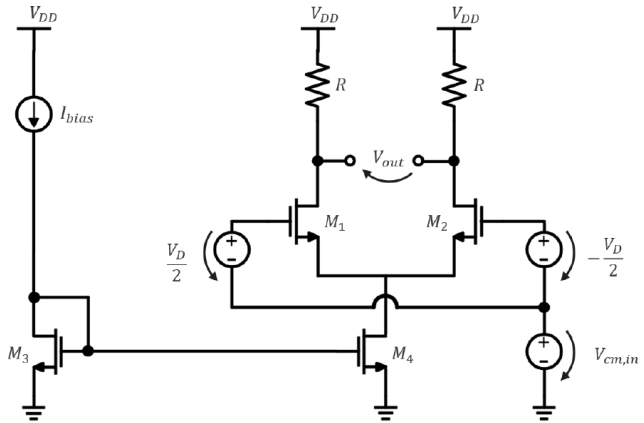
Exercise 1

i) Decide what length L_3, L_4 and V_{OV} you would choose for M_3 and M_4. Explain why? Then, calculate the width W_3 and W_4 with aid of Fig. 6.

$V_{Dsat} \approx V_{OV}$ (very pessimistic assumption)

to ensure saturation for all M_x:

$$V_{OV,3,4} + V_{OV,1,2} + \frac{V_{swing}}{2} \leq V_{DD}$$



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Exercise 1

Reminder $I_{SS} = 10I_{bias}$

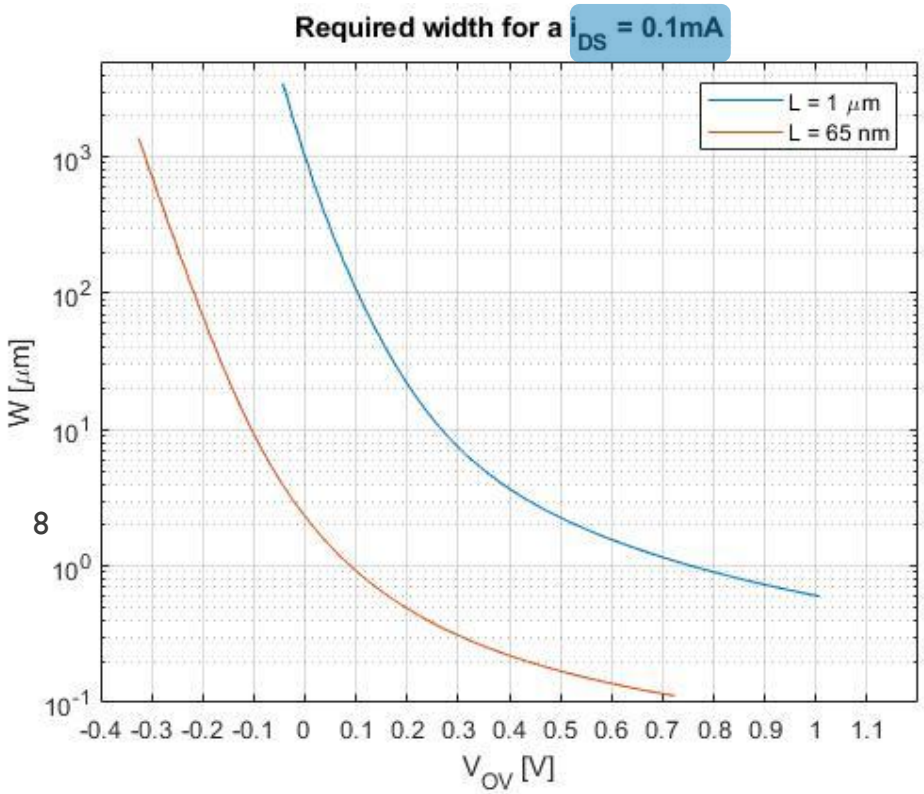
i) Calculate the width $W_{3,4}$

$I_{DS,4} = I_{SS}$
$=$

$W_4 \Rightarrow$

Then, with current mirror equation determine W_3

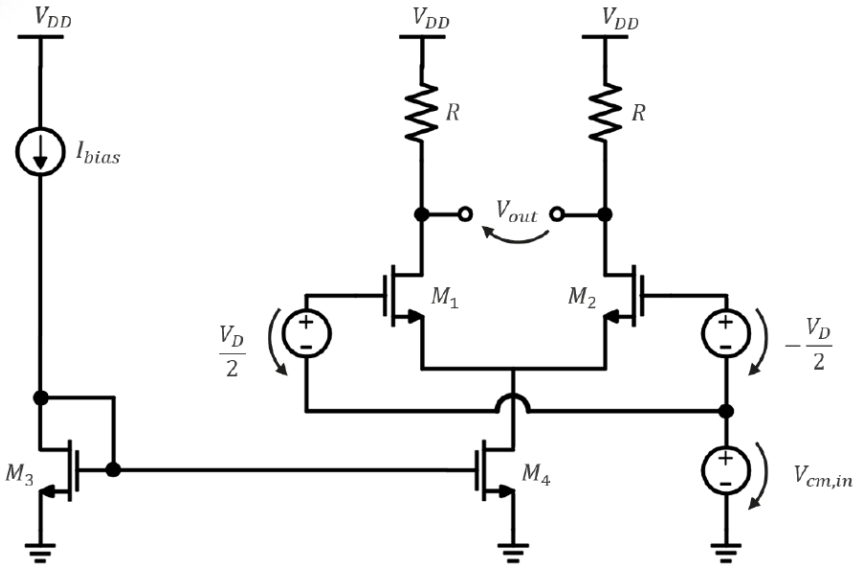
$\frac{I_{SS}}{I_{bias}} = \quad \Leftrightarrow W_3 =$



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Exercise 1

j) Calculate the minimum allowed input common mode voltage $V_{cm,min,in}$. What happens, if one sets $V_{cm,in} < V_{cm,min,in}$?



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Exercise 1

j) What happens, if $V_{cm} < V_{cm,in,min}$?

→ drives M4 into triode ....

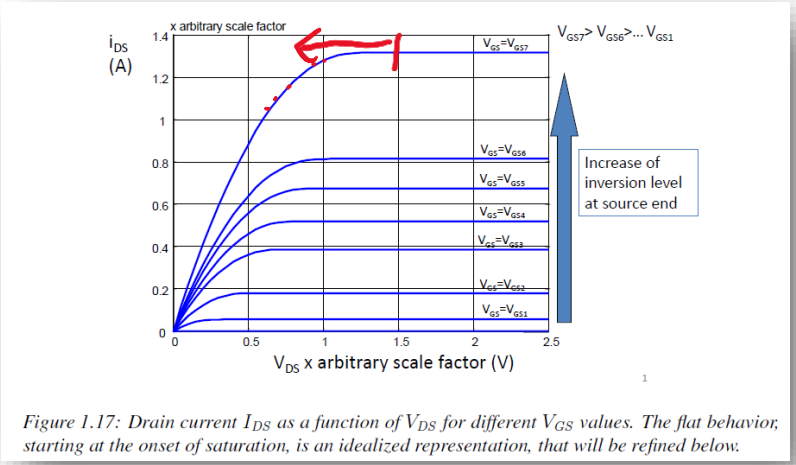
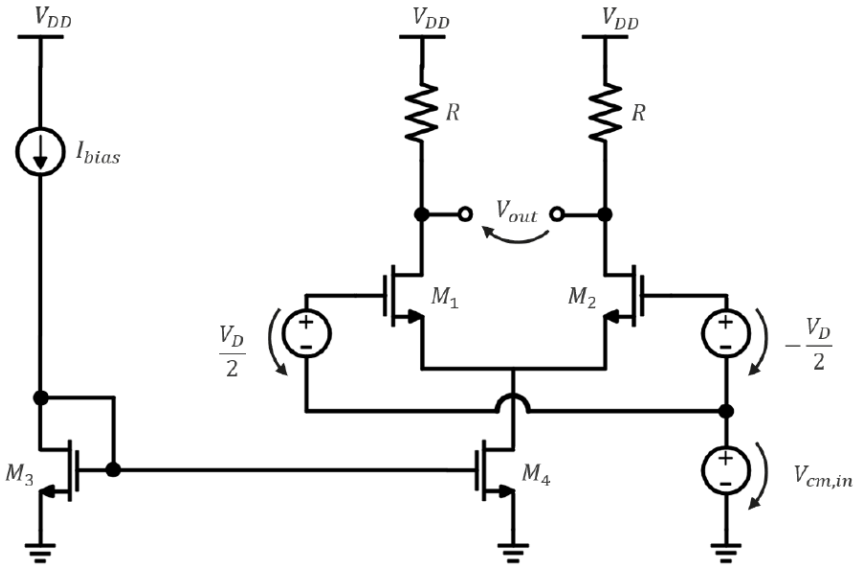


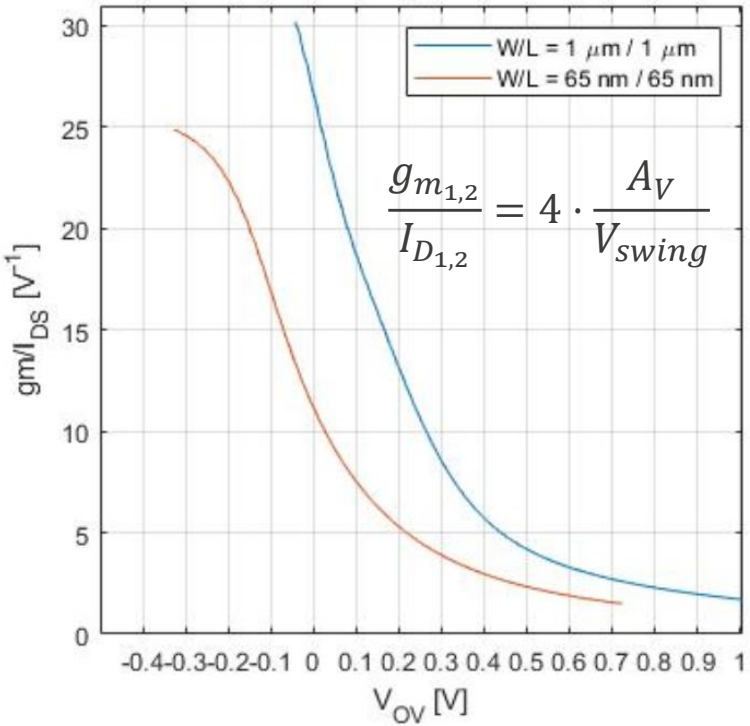
Figure 1.17: Drain current I_{DS} as a function of V_{DS} for different V_{GS} values. The flat behavior, starting at the onset of saturation, is an idealized representation, that will be refined below.



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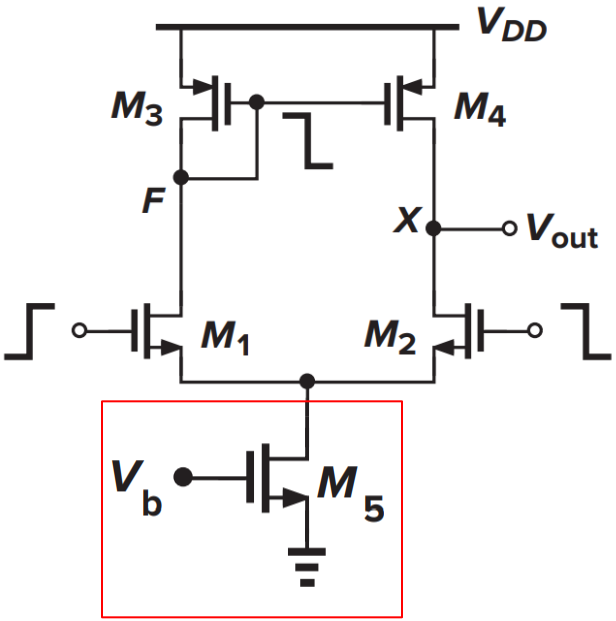
Summary

- Value the $\frac{g_m}{I_D}$-plot !
  - With $\frac{g_m}{I_D}$ one can identify the entire trade-off in one plot !



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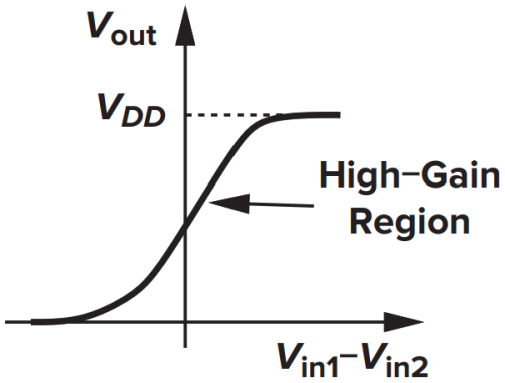
Differential Pair with Active Load



When the input at M1 increase, ID1 increases, VF falls, and ID2 decreases.

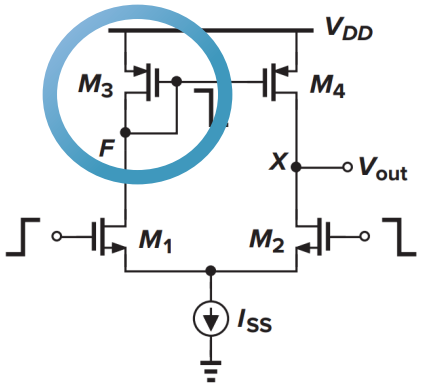
Thus, the output voltage rises by means of two mechanisms:

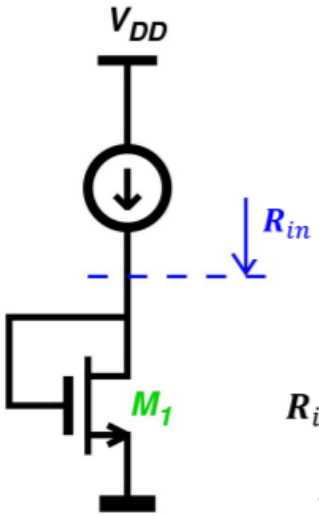
- M2 draws less current from X to ground and,
- M4 pushes a greater current from VDD to X.

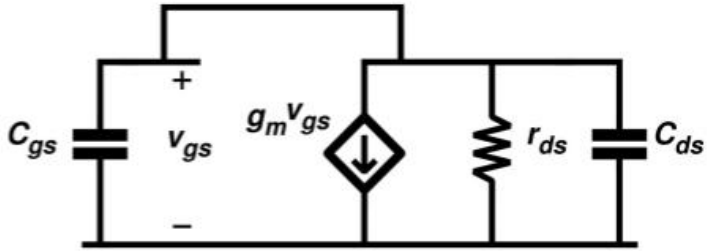


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Differential Pair with Active Load





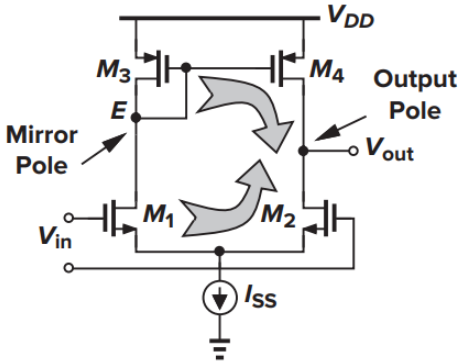


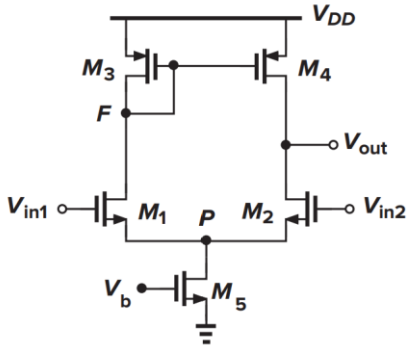
$$R_{in} = \frac{1}{g_m} || r_{ds} = \frac{1}{g_m + g_{ds}} \approx \frac{1}{g_m}$$

low impedance!

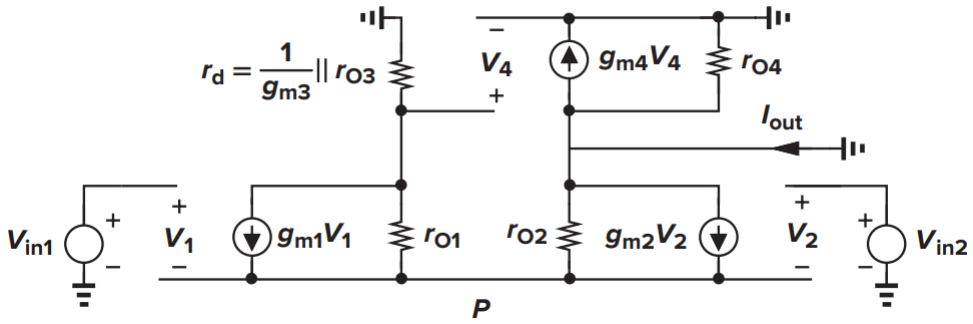
Session 3 : Design of differential pairs and of an OTA

Differential Pair with Active Load





Small signal equivalent



Hint!

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- Solve exercise 2 a-e
  - ETA: 30 min

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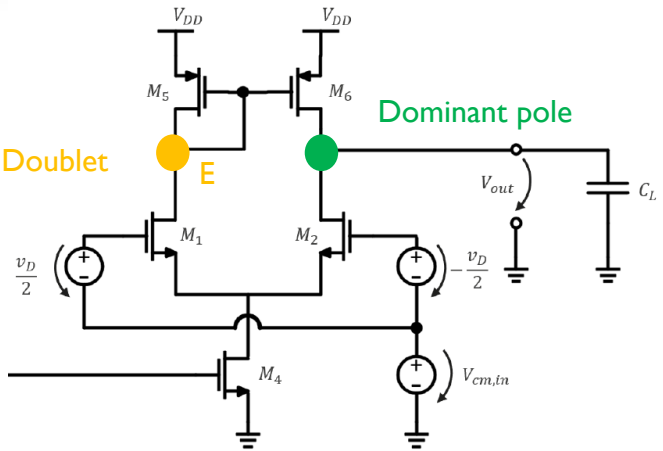
Exercise 2

a) Give the small signal voltage gain equation of this circuit including the capacitances and indicate

(i) dominant pole,

(ii) non-dominant pole,

(iii) zero.



Dominant pole

Non-dominant Pole

Zero

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Exercise 2: how to find the zero?

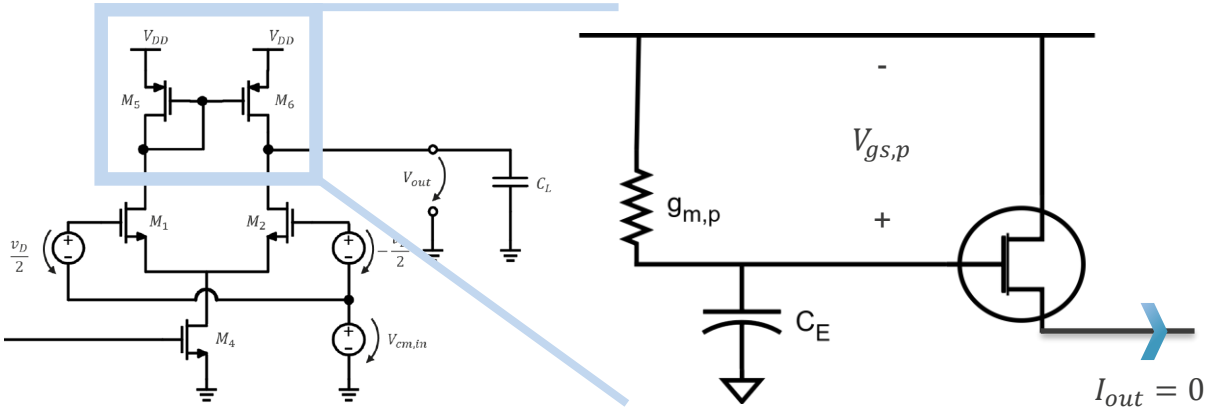
a) Give the small signal voltage gain equation of this circuit including the capacitances and indicate

(i) dominant pole,

(ii) non-dominant pole,

(iii) zero.

Equate the currents →



Circuit consists of a "slow path" (M1, M5, and M6) in parallel with a "fast path" (M1 and M2).

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Exercise 2

a) Give the small signal voltage gain equation of this circuit including the capacitances and indicate

   (i) dominant pole,

   (ii) non-dominant pole,

   (iii) zero.

b) Assume the pole-zero-doublet to be far away from the dominant pole. Draw the bode- and phase plot and indicate location of the dominant pole and GBW.

a)

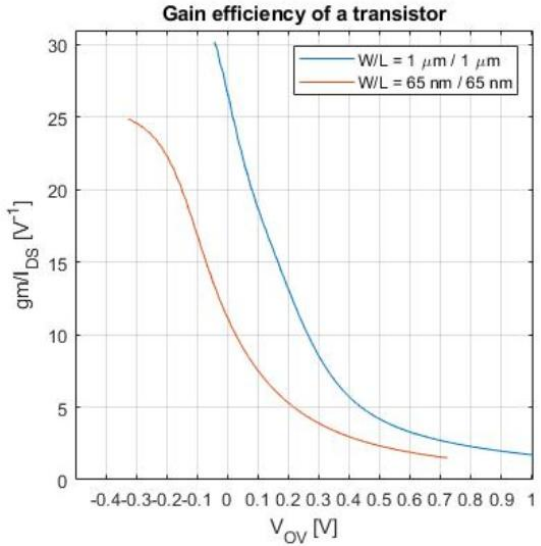
b) see drawing

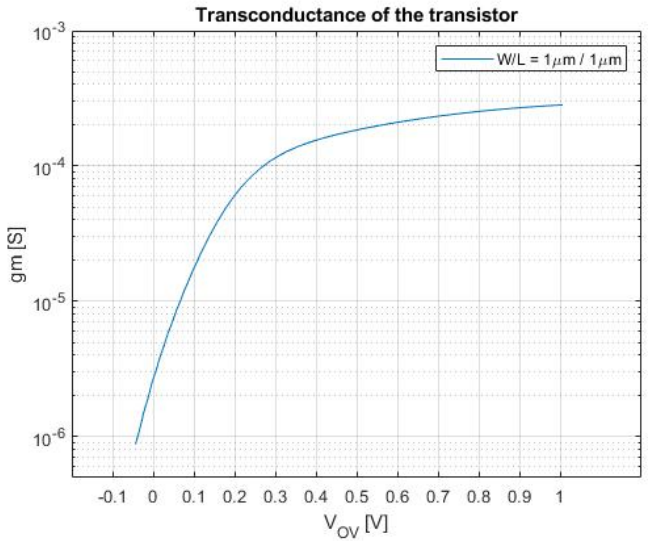
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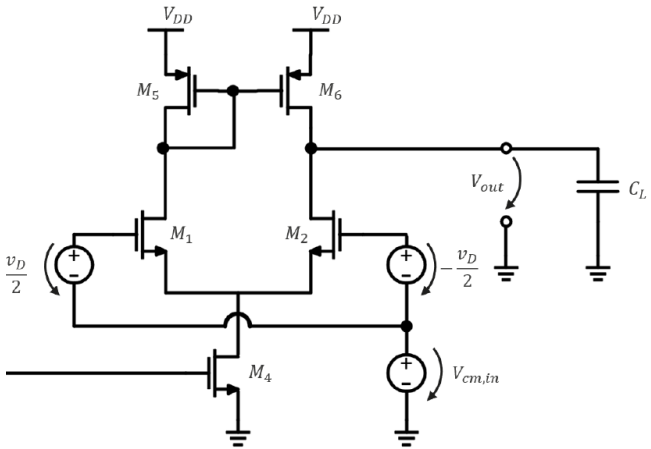
Exercise 2

c) What happens to the pole-zero-doublet, if one increases/decreases the V_{OV} of M_5 and M_6 assuming that the current flowing through M4 is fixed as well as the lengths are fixed?

c) $p_{non-dom} \sim \frac{g_{m5}}{C_{gs_{5,6}}}$ $V_{OV,5}$ g_{m5} $p_{nom-dom}$





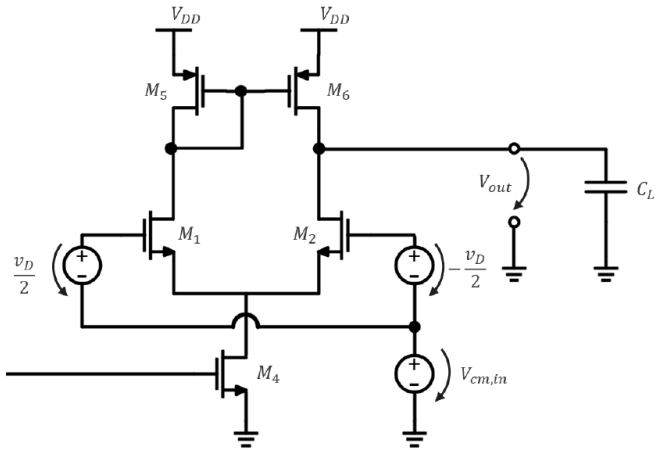


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Exercise 2

c) $p_{non-dom}$~

$V_{OV,5}$ $p_{nom-dom}$

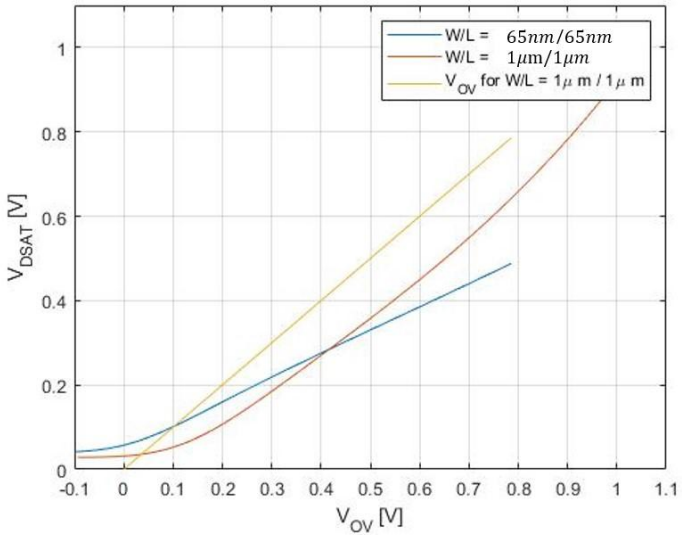


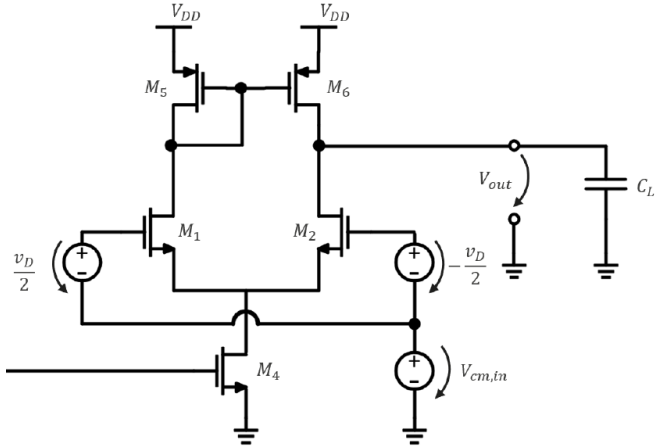
Session 3 : Design of differential pairs and of an OTA

Exercise 2

d) Assume $V_{OV,3,4} = 0.2V$, $V_{Dsat,5,6} = -0.25V$ and $V_{OV} < -0.1V$ for the remaining devices. With aid of Fig. 9, calculate the maximum/minimum possible output common mode voltage $V_{cm,max,out}$ and $V_{cm,min,out}$, where all transistors are still in saturation.

d) Calculate $V_{out,cm,min}$ (for $V_{OV,4} = 0.2V$ and remaining $V_{OV} < 0V$)



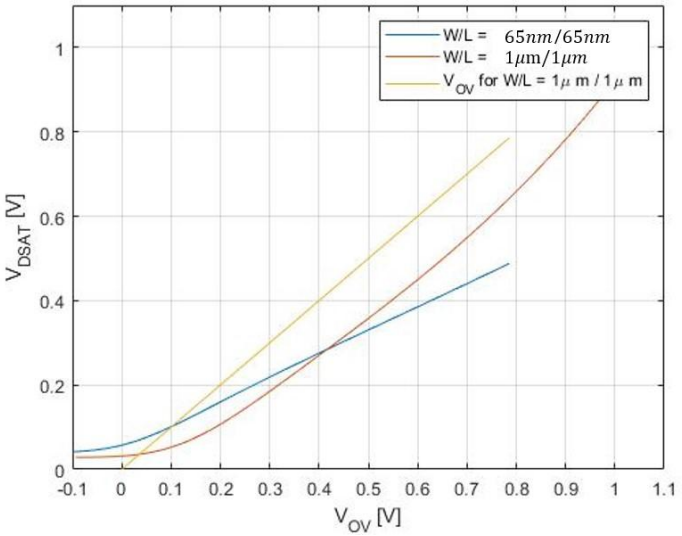


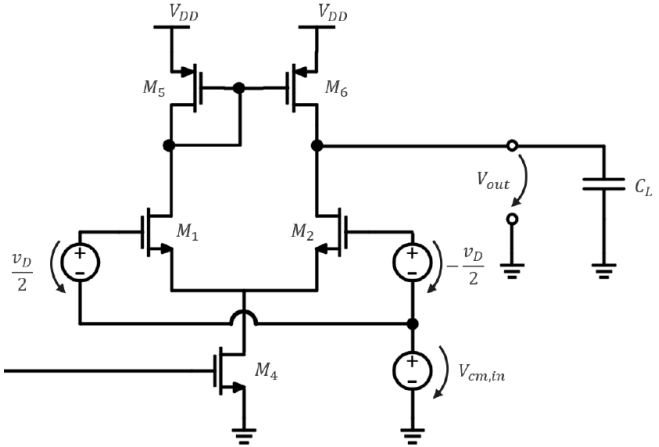
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Exercise 2

d) Calculate $V_{out,cm,max}$ (for $V_{OV,4} = 0.2V$ and $V_{Dsat,5,6} = -0.25\text{V}$)

$V_{out,cm,max} =$





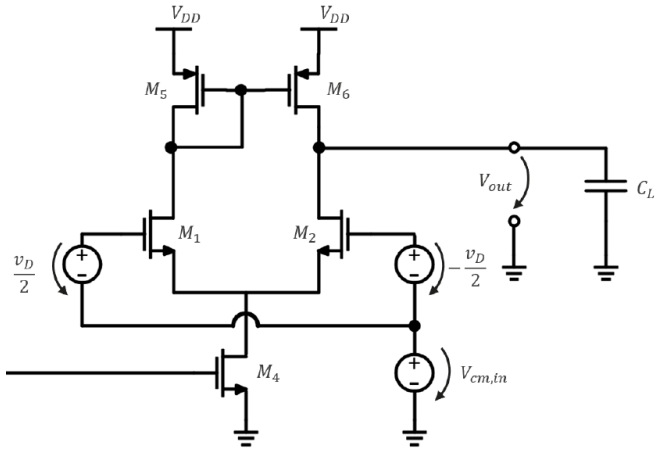
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Exercise 2

e) The designer decides to set the output common mode voltage for $v_D = 0$ to the midpoint between $V_{cm,max,out}$ and $V_{cm,min,out}$. Explain how and calculate for that the required V_{OV}.
Hint: Examine the PMOS pair for $v_D = 0$.

e) $V_{out,cm}$ calculated as follows

$V_{out,cm} =$

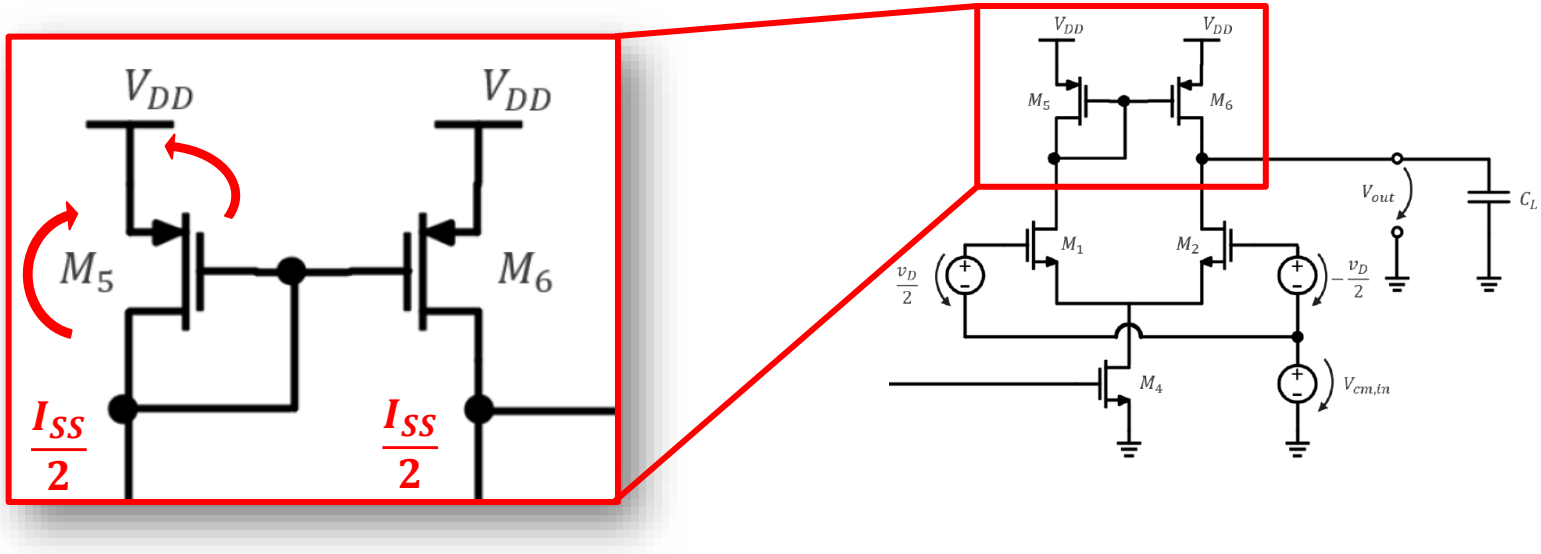


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Exercise 2

e)

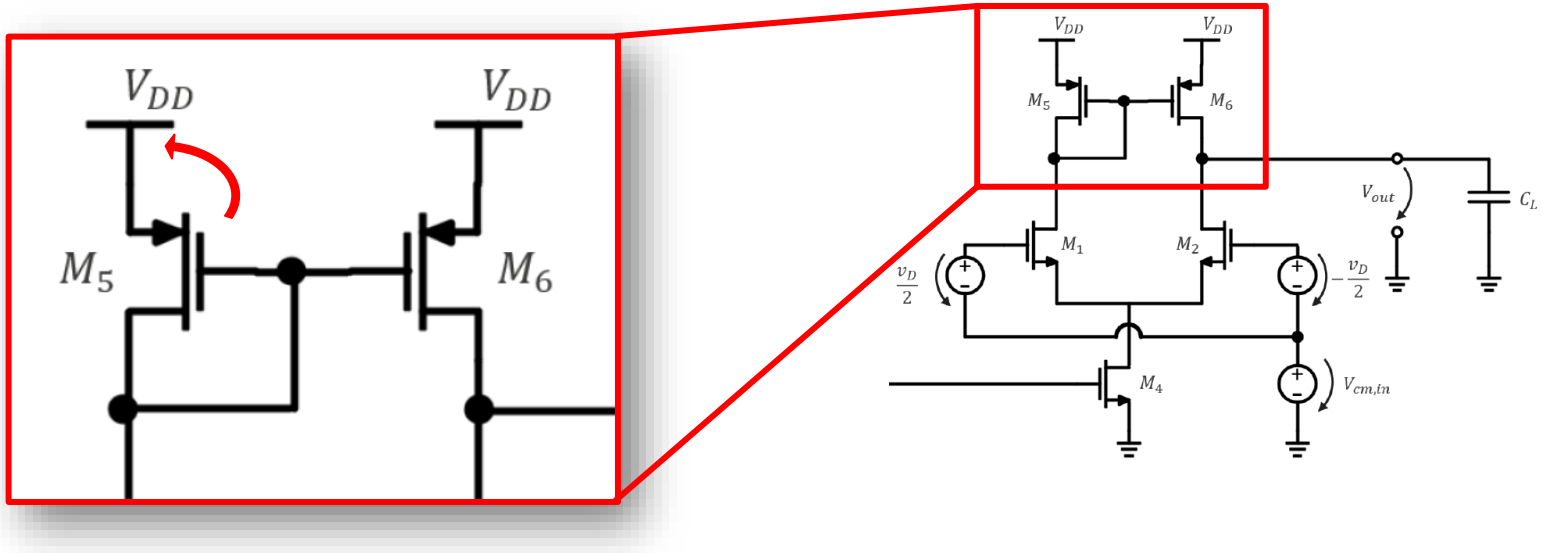
How to setup $V_{out,cm}$ in OTA's ?



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Exercise 2

e) Consequently, $|V_{OV,5}| =$



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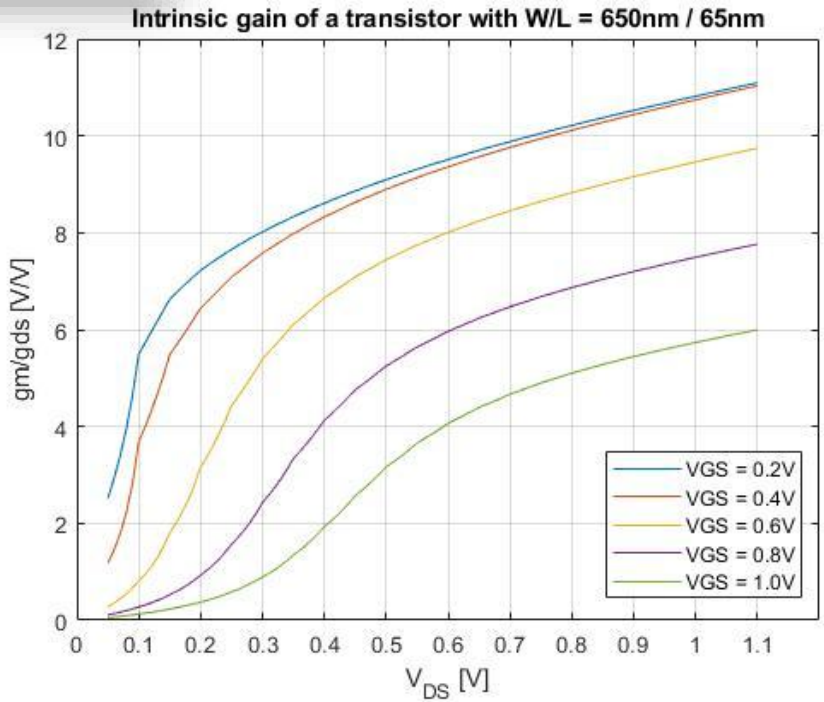
- Solve exercise 2 f-i
  - ETA: 20 min

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Exercise 2

f) What is the expression for A_v in low frequencies (below the dominant pole frequency)? Use Fig. 8 to calculate required V_{OV} and $V_{in,cm,min}$ for M1 and M2 to achieve $A_v > 8$.

f) How to simplify $A_V(s = 0)$?



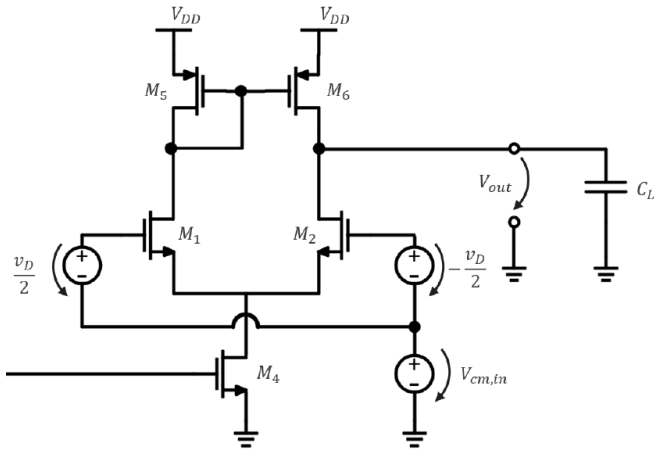
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Exercise 2

f) $V_{DS,2}$ calculated as follows

$V_{DS,2} =$



Session 3 : Design of differential pairs and of an OTA

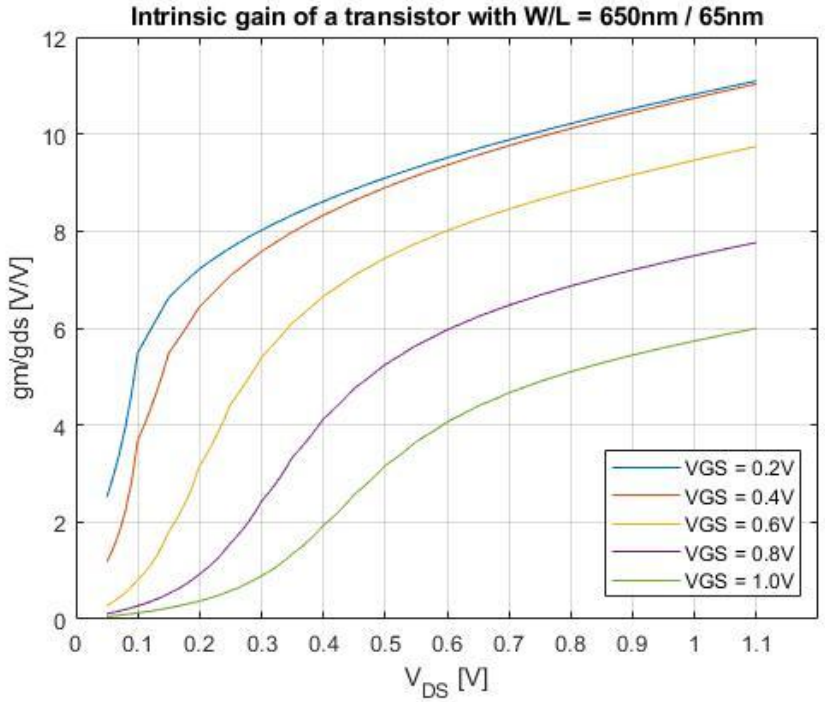
Exercise 2

f) Search $V_{OV,1,2}$ for $A_V(s = 0) > 8$

→ $V_{GS,1,2} =$

→ $V_{OV,1,2} =$

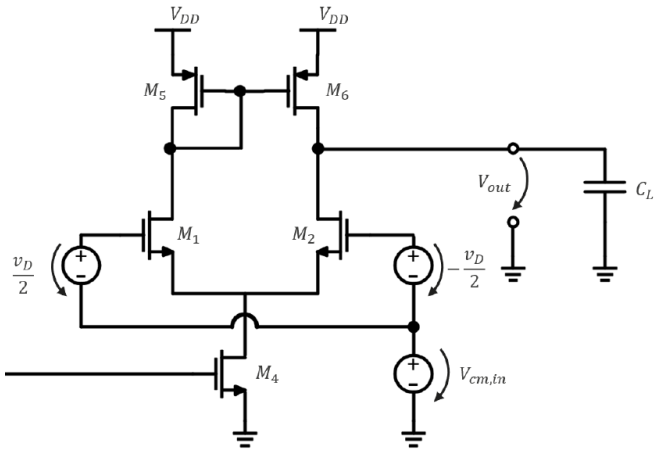
→ $V_{in,cm} =$



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Exercise 2

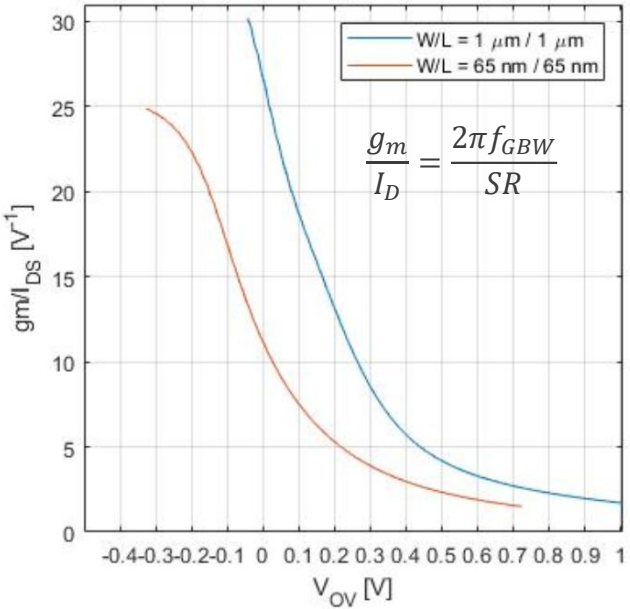
g) Calculate $V_{in,cm,max}$.

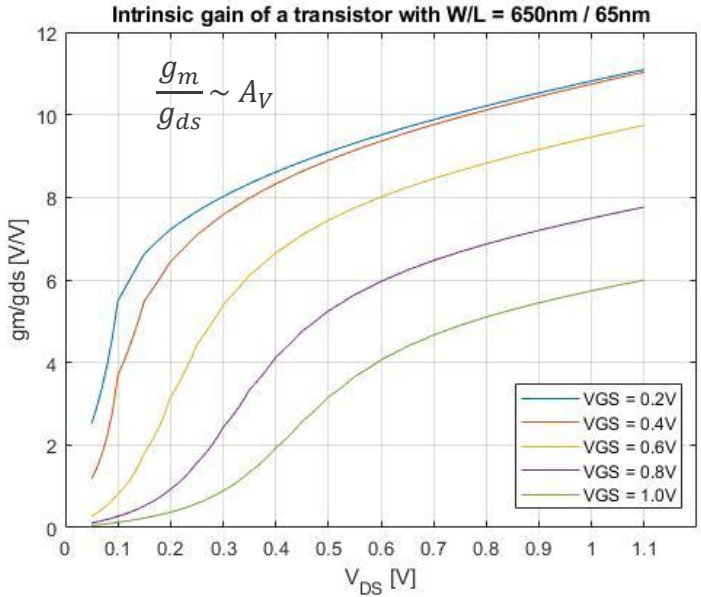


Session 3 : Design of differential pairs and of an OTA

Summary (1)

1. Value the $\frac{g_m}{I_D}$- and $\frac{g_m}{g_{ds}}$- plots!
   - ➢ One can predict the feasibility of the given specifications for an OTA in one view

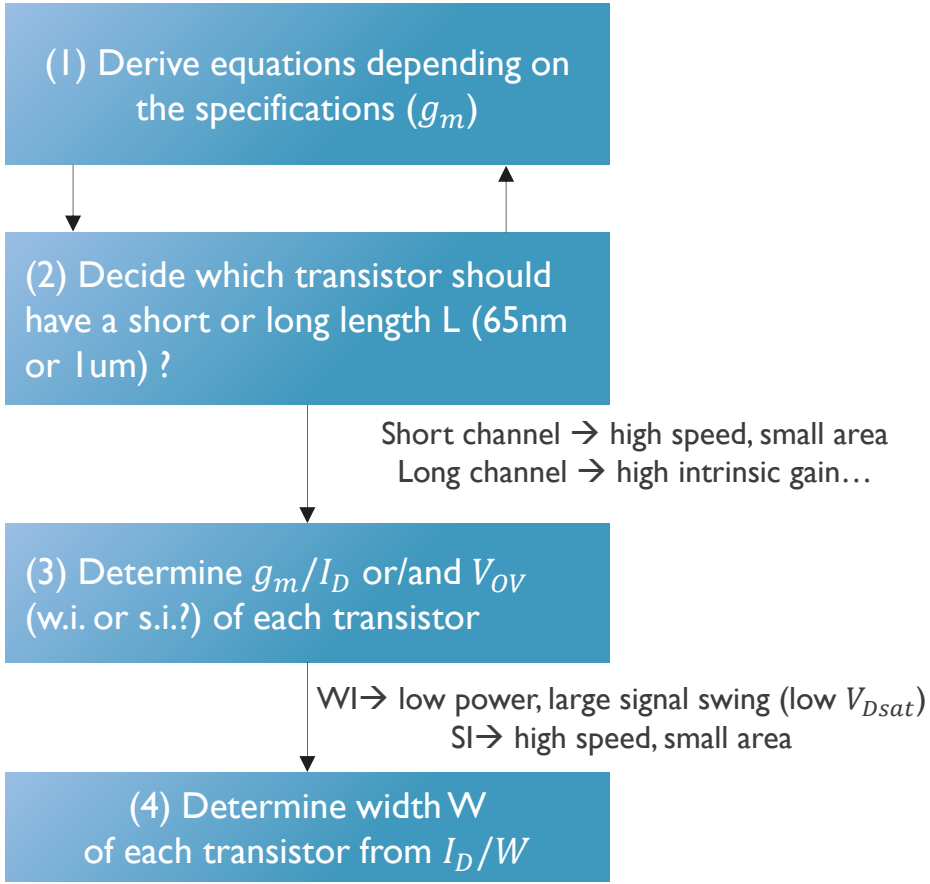




Session 3 : Design of differential pairs and of an OTA

Summary (2)

2. Design flow introduced
3. Learned how to "design on paper" with
   - W-plot for fixed g_m or i_{ds}
   - $\frac{g_m}{I_D}$- and $\frac{g_m}{g_{ds}}$-plots
   - V_{dsat}-plot
   - ➢ Those plots will help you to save time when designing for the final lab session!



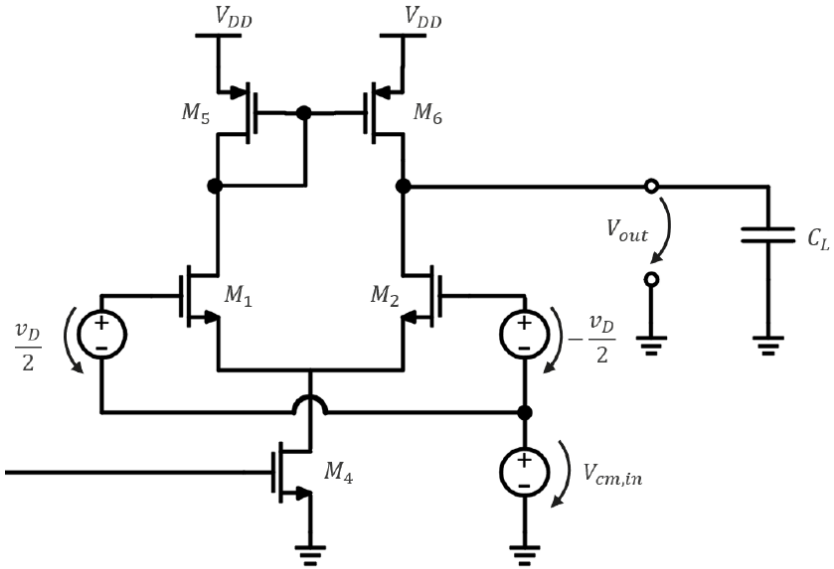
Session 3 : Design of differential pairs and of an OTA

MATLAB session

- Exercise 3
  1. Type in the code accordingly to this presented OTA exercise
  2. Solve following scientific questions?
     1. How can one move the pole-zero-doublet?
        - 2 options
     2. How can I increase the f_{GBW}?
  - ETA: 30 min

Plug in the values below!

```
M4.ids    = 200e-6;
SR          = M4.ids/spec.CL/1e6 % [mV/ns] slew rate
```



Hints:

1. Use the following command to copy transistors e.g. M1 and M2
   1. M2 = cirElementCopy(M1, M2);
2. Before coding, draw all voltage condition on the OTA-circuit
   - i.e. vgs, vgd, vds, ids (or gm) of each transistor !
   - This will help you to speed-up your coding
3. Code it smart !
   1. code it such that you can play around with the pole-zero doublet !

Session 3 : Design of differential pairs and of an OTA

... some helpful references

[1]: “Analog Design Essentials”, Sansen, Willy M. C, 2006

[2]: “Design of Analog CMOS Integrated Circuits”, Razavi, Behzad, 2016

[3]: “Analysis and Design of Analog Integrated Circuits”, P. Gray, P. Hurst, S. Lewis and R. Meyer, John Wiley and Sons, 2009


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